

1. Week 8 — Ethics & Real-Life Scenarios

Key Ethics Codes to Remember

- **ACM Code** — Honor contracts, agreements, and assigned responsibilities
- **Fairness & Discrimination** — Treat everyone equally; no favoritism or bullying
- **Security & Unauthorized Access** — Never give unauthorized access to data
- **Cyber Bullying** — Threatening or harassing others online is illegal

Case-by-Case Summary

Case	Scenario (in short)	Ethics Code Violated	Correct Response
Case 2	Ahmed used 3 teammates' code without giving them credit; also added friend Rashid's code without telling management	**ACM Code** — Not giving credit, lack of transparency	Always acknowledge others' contributions ; inform management of all external code
Case 3	Canadian software house delivered tax software with wrong formulae; clients got penalized by Tax Dept	**ACM Code** — Did not honor contract	Thoroughly test software before delivery; do not sign false contracts
Case 5	CEO asks junior developer to spy on project manager in exchange for benefits	Fairness & Discrimination	Refuse the offer politely; it is unethical to spy on your boss; follow proper organogram

2. Week 9-10 — Human Resource Management (HRM)

What is HRM?

- **HRM** = Managing employees so their performance is **maximized** according to the company's **strategic objectives** (aims & goals)
- Very important in **software development** organizations

HRM Responsibilities

- Organization has **expectations** from employees → employees must manage themselves and their work
- HRM is the **responsibility of management**
- Management ensures **maximum utilization** of human resources

Three Levels of Management

Level	Also Called	Who?	What They Do
Strategic	Top Level	Chairman, MD, CEO, Board of Directors (BOD)	Develop organizational goals & strategic objectives (e.g., new branch, new product)
Tactical	Middle Level	Department Heads, Managers	Implement strategies set by top management
Operational	Front-line	Supervisors, Team Leads	Day-to-day operations & direct employee supervision

3. Week 11-12 — Software Liability

Types of Bugs

- **Out of Bound Bugs** — User enters a value **beyond the allowed limit** (too large, too small, wrong data type)
- **Security Bugs** — Developer adds security restrictions but they **don't actually work** (e.g., password check is bypassed)

Operating System Failures

An OS failure = computer gives **strange messages** or **hangs** while running applications.

Cause Type	Examples
Software Problems	OS not properly installed; malware/virus infection; unauthentic internet downloads
Hardware Problems	Damaged storage device; faulty power supply (sudden shutdown); bad processor; incorrect RAM chip

OLTP (Online Transaction Processing)

- **Definition:** A data processing system that executes **many transactions over the internet**
- **Examples:** Online banking, shopping, order entry, admissions, job hunting, messaging

OLTP Performance Issues

- Must handle **hundreds/thousands of simultaneous users**
- Must allow **many users to work on the same data** with **immediate updating** (e.g., airline reservations)
- Must handle **errors properly**

Distributed vs Centralized Databases

Feature	Distributed Database	Centralized Database
Data Location	Multiple files at **different sites**	Single file at **one central location**
Multi-user Access	Allows multiple users to access & manipulate data **simultaneously**	**Bottleneck** when many users access same file simultaneously
Delivery Speed	Files delivered **quickly** from the nearest location	Files **may take longer** to deliver
Fault Tolerance	If one site fails, **data is still retrievable** from other sites	Single site failure = **downtime for everyone**

Real-World Examples of Distributed Databases

- **Facebook** messages → uses **Apache HBase** (streams data to Hadoop clusters)
- **YouTube** → uses **Kubernetes-aware** cloud-native distributed database
- **TikTok** (ByteDance) → uses **ByteGraph** (distributed graph database)

4. Week 15 — Cashflow Quadrant

The ESBI Quadrant — 4 Ways to Produce Income

	Works for others	**Works for self / owns**
Time = Money (income stops when you stop)	**E — Employee**	**S — Self-Employed**
System/Money works for you (income continues)	**B — Business Owner**	**I — Investor**

Detailed Breakdown

Quadrant	Description	Key Point	Example
E — Employee	Has a J.O.B., works for someone else	Income **stops** when work stops; financial destiny depends on **employer's success**	Low, Mid, Top level employees
S — Self-Employed	Own boss, works for themselves	More **control** than E, but still **trades time for money** ; leads to tiredness & fatigue	Dentists, lawyers, restaurant owners, insurance/property agents
B — Business Owner	Owns a **system** , hires people to run it	Earns money **without constant involvement** ; system runs automatically	Law firm owner (juniors plead cases), franchise owners
I — Investor	Owns **assets** that produce income	**Passive income** ; greatest financial freedom; money works for them	Shareholders, property investors, business traders, influencers

Famous Investment Quotes

- **"Investing in yourself is the best investment you will ever make."** — **Robin Sharma**
- **"The more you learn, the more you earn."** — **Warren Buffett**
- **"Your career is the engine of your wealth."** — **Warren Buffett**
- **"Given a 10% chance of 100x payoff, you should take that bet every time."** — **Jeff Bezos**

5. Week 15-16 — CV vs Resume

Cover Letter (Opening of CV/Resume)

A cover letter has **4 paragraphs**:

Paragraph	Content
1st	How you **came to know** about the job
2nd	Your **experience and skills**
3rd	How your skills **match the job** requirements
4th	Express your **wish to meet** the interviewer

What is a CV?

- **CV** = Curriculum Vitae = describes your **whole career**
- Length: Usually **2-3 pages**, can be **10+ pages** if needed
- Used for **academic purposes**
- Presents: education, professional career, publications, awards, etc.

CV Sections (Full List)

- Contact information
- Research objective / personal profile / personal statement
- Education
- Professional academic appointments
- Books & Book chapters
- Peer-reviewed publications & Other publications
- Awards and honors
- Grants and fellowships
- Conferences
- Teaching experience
- Research / lab experience / graduate fieldwork
- Non-academic activities
- Languages and skills
- Memberships
- References

What is a Resume?

- A **short, concise** document for **job applications**
- Length: **1-2 pages** only
- **Targeted** at a specific job — only includes **relevant** details
- Purpose: give recruiters a **brief overview** of work history

Resume Sections

- Contact information (including job title)
- Resume summary or objective
- Work experience
- Education section
- Skills section
- Additional details (awards, courses, certifications, personal interests)

CV vs Resume — Quick Comparison

Feature	CV	Resume
Length	No limit (2-3 to 10+ pages)	**1-2 pages** only
Content	**Entire career** history	Only **relevant** experience & skills
Purpose	**Academic** (research, teaching, fellowships)	**Job applications**
Customization	Same CV for all applications	**Tailored** to each specific job

Resume Do's

- Keep it **1-2 pages**
- **No picture** unless applying for: brand ambassador, sales, marketing, receptionist, anchoring, modeling
- Summary statement should include:

- **Soft skills:** enthusiastic, passionate, hard-working
- **Technical skills:** industry experience, strengths
- Education listed in **reverse chronological order** (newest first)

6. Week 16 — Continuing Professional Development (CPD)

What is CPD?

- **CPD** = The ongoing process of **identifying professional goals**, **developing a plan** to achieve them, and **documenting** the experience
- It is a **cycle** — it never stops (lifelong learning)

The 5 Stages of the CPD Cycle

Stage	Name	What to Do	Key Action
1	**Identify Needs**	What is your goal? Summarize your career path, imagine milestones, build a **roadmap** of small achievable goals	Define your destination
2	**Planning**	Identify gaps & growth areas; set **SMART goals** ; outline activities & resources needed	Map the path
3	**Action**	Attend training, coaching, certificate courses, webinars, conferences; use platforms like **Udemy, Coursera, Udacity, Google free courses**	Start learning
4	**Reflect**	Consider what you learned and how it influenced your work; **write down** observations; identify areas for further improvement	Look back & evaluate
5	**Implement**	Apply your new learning; promote internal training; request online courses; **never stop** after 1-2 certificates — participate in **lifelong learning**	Put it into practice

SMART Goals (with Example)

SMART = Specific, Measurable, Attainable, Relevant, Time-bound

Example: Learning English for Freelancing

Letter	Meaning	Example
S — Specific	Clearly define what you want	**"I want to learn conversational English for client communication"
M — Measurable	How will you track progress?	**"I will use a daily language app to track progress"
A — Attainable	Is it realistically achievable?	**"I know English speakers on social media to practice with"
R — Relevant	Does it matter to your career?	**"Most of my freelance clients are native English speakers"
T — Time-bound	Set a deadline	**"I want to learn in the next 3 months"